

MU60x Series Modules

Host Interface Packet Definitions

Document summary

Document name	Classification
MU60x Host Interface Packet Definitions	open

Version history

Edition	Date	Change description
V1.0.0	2022.09.08	Initial version
V1.0.1	2022.10.26	Modify the firmware format to add model identification Modify the antenna serial number. The minimum is 0x01 Modify the frequency setting format to add one byte as the frequency interval Modify the RF link options Modify the tag reply command to RSSI and Freq length
V1.0.2	2022.12.03	The repair read/write protocol reply format is incorrect
V1.0.3	2022.12.08	The document was repaired without stopping the inventory command
V1.0.4	2022.12.14	The length of the modified read protocol is incorrect
V1.0.5	2022.12.22	Add SM7 related commands
V1.0.6	2023.02.17	Add a custom inventory command and change the RSSI calculation formula
V1.0.7	2023.03.16	Modify the inventory command report format and delete the tag filter command
V1.0.8	2023.04.04	Error code descriptions are added to individual instruction parsing
V1.0.9	2023.04.12	Added an antenna switching inventory command
V1.1.0	2023.05.05	Add new national standard orders, add new Fudan Micro national standard orders, and modify the error response of some orders
V1.1.1	2023.06.26	Added 4, 8 channel reader output power setting command; optimized multi-antenna polling inventory description
V1.1.2	2023.07.17	Modify the packet definition description and restore the default parameter instruction description Update the instruction set list Add antenna detection command, over temperature alarm
		command and parameter saving command Modify the setting of the work antenna instruction data packet description Add section 8 Default parameter configuration table for factory

V1.1.3	2023.11.30	The document has been reorganized Add partial command reference examples Modify the length of the 8 antenna read to get the current output power Modify the note information of the RF output power of a single antenna reader Modify the data packet format definition checksum Note rest Modify the query inventory cache to return cmd command Added new CW related commands Added Session and Target commands Add new inventory heartbeat packets and report related commands Added TX ON and TX OFF related commands Add a new selection tag related command
V1.1.4	2024.2.29	The antenna detection related commands have been cancelled
V1.1.5	2024.3.11	Increase the number of read working antennas-multiple antennas Modify the note of Data field in 80H inventory tag Modify the note of the Data field in the 89H inventory tag (real-time) Modify the 8AH inventory tag (custom) Data field note Add 8AH inventory tag (custom) matching parameter related remarks command description
V1.1.6	2024.5.15	Added GPIO Settings and read commands
V1.1.7	2024.6.24	Added the read UUID command
V1.1.8	2024.7.12	The CRC check return for the tag when the 8A command setting is not matched has been removed
V1.1.9	2024.9.2	Modify the checksum description and add the LRC check description
V1.2.0	2025.2.14	Modify the description of the range of operating frequencies for the reader.
V1.2.1	2025.3.3	correct errors in printing

preface

This document is the definition of the serial communication protocol for ultra-high frequency RFID reader modules, primarily intended for developers using the series of ultra-high frequency RFID reader modules. The protocol specifies the communication format between the reader and the host computer or other control platforms. Developers can follow this document to develop their applications while retaining the original firmware of the reader or using a secondary development version of the reader while keeping the existing serial protocol.

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1. Document overview

1.1. Name

MU60x Host Interface Packet Definitions.

1.2. Description

This protocol specifies the communication format between the UHF RFID reader module and the host computer.

1.3. Application score

MU60x Series modules.

2. Packet format

2.1. Host computer data packet structure

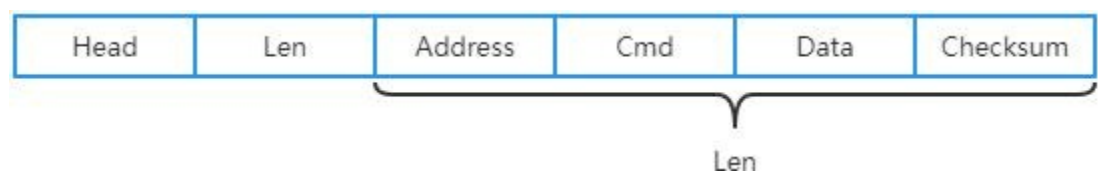


Figure 1 Packet format sent by the host computer

NO.	Field name	NO.of bytes (Byte)	explain
1	Head	1	Packet header, each packet starts with 0xA0
2	Len	1	The packet length does not include the packet header and the packet length itself. If the complete packet length is 8 bytes, the value of Len is 6
3	Address	1	Reader address. Used for RS-485 interface series connection. The general address is from 1 to 25 5 (0xF F), and 0 (0x 00) is the common address. The reader receives

			commands from its own address and the common address
4	Cmd	1	Command code, an enumeration value defined in RFID_COMMAND
5	Data	0~N	Command parameters may be 0 bytes or multiple bytes
6	Checksum	1	LRC verification, The checksum of all bytes except the check field itself is taken modulo 0x100, and then 0x100 is subtracted from the modulo (or the checksum is inverted and added to 1) C routine <pre>static rt_uint8_t checksum (rt_uint8_t* buff, rt_uint32_t data_len) { rt_uint32_t index= 0; rt_uint16_t sum= 0; for (index= 0; index<data_len; index++) { sum= sum+ buff[index]; } return ((0x100-(sum% 0x100))& 0xff); }</pre>

2.2. Data packet format returned by reader

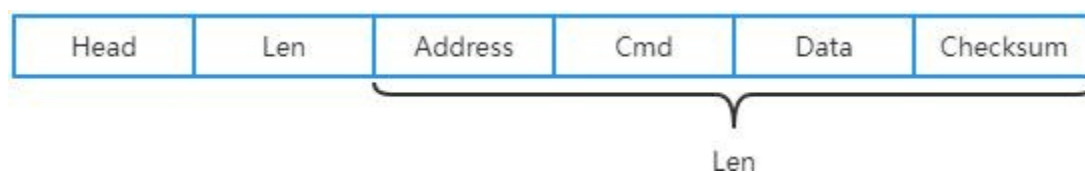


Figure 2 Format of data packet returned by the reader

NO.	Field name	NO.of bytes (Byte)	explain
1	Head	1	Packet header, each packet starts with 0xA0
2	Len	1	The packet length does not include the packet header and the packet length itself. If the complete packet length is 8 bytes, the value of Len is 6

3	Address	1	The address of the reader itself
4	Cmd	1	Command code, an enumeration value defined in RFID_COMMAND
5	Data	0~N	The data returned by the reader may be zero bytes or multiple bytes
6	Checksum	1	LRC verification, The checksum of all bytes except the check field itself is taken modulo 0x100, and then 0x100 is subtracted from the modulo (or the checksum is inverted and added 1) C routine <pre>static rt_uint8_t checksum (rt_uint8_t* buff, rt_uint32_t data_len) { rt_uint32_t index= 0; rt_uint16_t sum= 0; for (index= 0; index<data_len; index++) { sum= sum+ buff[index]; } return ((0x100-(sum% 0x100))& 0xff); }</pre>

Note: Unless otherwise specified, the Data segment in the data packet returned by the reader is an error code. Please refer to Chapter 4--Error Code Table for the specific error code.

2.3. Instruction set summary

NO.	comm and code	name	description
1	0x3E	cmd_set_cw	Turn on the cw wave
2	0x3F	cmd_get_cw	Get the cw wave status
3	0x42	cmd_gb_seu_write	Fudan Micro GB encryption communication write
4	0x43	cmd_gb_seu_read	Read Fudan Micro GB encrypted communication
5	0x45	cmd_gb_read	Read the tag GB
6	0x46	cmd_gb_write	Write the tag GB
7	0x47	cmd_gb_lock	Lock the tag GB
8	0x49	cmd_gb_kill	Inactivate the tag GB
9	0x4a	cmd_reader_para_save	Save parameters

10	0x4b	cmd_reader_para_reset	Restore the default configuration of the reader
11	0x4c	cmd_reader_app_upgrade	Reader application software upgrade
12	0x4d	cmd_baseband_firmware_upgrade	Baseband firmware upgrade
13	0x4E	cmd_set_keepalive	Set the heartbeat packet and report time
14	0x4F	cmd_get_keepalive	Get heartbeat packets and report time
15	0x5A	cmd_get_session_target	Get session and target
16	0x5B	cmd_set_session_target	Set session and target
17	0x5C	cmd_get_tx_time	Get the tx_on time and tx_off time of the reader
18	0x5D	cmd_set_tx_time	Set the tx_on time and tx_off time of the reader to adjust the power consumption
19	0x5E	cmd_set_8_ant_power	Set the 8 antenna reader RF output power
20	0x5F	cmd_set_4_ant_power	Set the 4 antenna reader RF output power
21	0x60	cmd_read_gpio_value	Query the GPIO status
22	0x61	cmd_set_gpio_value	Set GPIO status
23	0x66	cmd_set_temporary_output_power	Set the temporary radio frequency output power of the reader
24	0x69	cmd_set_rf_link_profile	Set the communication rate of the RF link
25	0x6A	cmd_get_rf_link_profile	Query the communication rate of the RF link
26	0x70	cmd_reset	Reset the reader
27	0x71	cmd_set_uart_baudrate	Set the serial port communication baud rate
28	0x72	cmd_get_firmware_version	Query the firmware version of the reader
29	0x73	cmd_set_reader_address	Set the address of the reader
30	0x74	cmd_set_work_antenna	Set the working antenna of the reader
31	0x75	cmd_get_work_antenna	Query the current working antenna
32	0x76	cmd_set_output_power	Set the RF output power of the single antenna reader
33	0x77	cmd_get_output_power	Query the current output power of the reader
34	0x78	cmd_set_frequency_region	Set the operating frequency range of the reader
35	0x79	cmd_get_frequency_region	Query the operating frequency range of the reader
36	0x7B	cmd_get_reader_temperature	Query the operating temperature of the current device
37	0x80	cmd_inventory	inventory tag
38	0x81	cmd_read	Read the tag

39	0x82	cmd_write	Write the tag
40	0x83	cmd_lock	Lock the tag
41	0x84	cmd_kill	Inactivated tags
42	0x87	cmd_fast_switch_ant_inventory	Multi-antenna polling inventory
43	0x89	cmd_real_time_inventory	Inventory tag (upload tag data in real time)
44	0x8A	cmd_custom_inventory	Inventory tags (custom read data)
45	0x8B	cmd_customized_session_target_inventory	Customize session and target inventory
46	0x8C	cmd_stop_inventory	Stop the inventory
47	0x8D	cmd_set_select	Set the select parameter
48	0x8E	cmd_get_select	Get the select parameter
49	0x90	cmd_get_inventory_buffer	Query inventory cache
50	0x91	cmd_get_and_reset_inventory_buffer	Query and reset the inventory cache
51	0x92	cmd_get_inventory_buffer_tag_count	Query the number of inventory cache tags
52	0x93	cmd_reset_inventory_buffer	Reset inventory cache
53	0x95	cmd_sm7_write	SM7 encryption write
54	0x96	cmd_sm7_read	SM7 decryption read
55	0x97	cmd_sm7_pk_update	SM7 PK Key update
56	0x98	cmd_gb_mul_SeU_Auth	Fudan Micro GB two-way authentication
57	0xE0	cmd_check_ant	Get the antenna connection status
58	0xE1	\	The temperature of the reader is too high
59	0xF3	cmd_get_uuid	fetch UUID

3. Instruction format

3.1. 3EH—— Set the CW switch status

Command: cmd_set_cw

The host computer sends instruction data

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x3E	
5	Data	1		0x1 Open CW 0x0 Turn off CW
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 3E 01 1D

Data analysis: 01: Open CW

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x3E	
5	Data	1		0x10, Settings are successful 0x11 Failure
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 3E 10 0E

Data analysis: 10: Settings are successful

3.2. 3FH—— Get the CW switch status

Command: cmd_get_cw

The host machine sends instruction data

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x3F	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 3F 1E

Data analysis: None

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x3F	
5	Data	1		The 0x1 CW is on 0x0 CW is off

6	Checksum	1		
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Reference example:

Serial port data: A0 04 00 3F 01 1C

Data analysis: 01: CW is open

3.3. 42H—— GB-standard encryption communication (write) for Fudan Micro tags

Command: cmd_gb_seu_write (0x42) Before executing this command, you must perform a two-way authentication!

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x42	
5.1	Data	Password	4		Enter the password for the tag
5.2		MemBank Tag storage area	1		0 tag information area 1 Coding area 2 Safe zone 3 User Area
5.3		WordAdd Data first address	2		WORD (16 bits) address. Writing into the EPC storage area usually starts from 01, and the first two bytes of this area store the PC.
5.4		WordCnt Write the length of data	2		WORD (16 bits) length, refer to the tag specification for the value
5.5		WordData	WordCnt *2		The data written (variable length) is twice the length of the data written.
6	Checksum		1		

The reader returns the data packet

Returns the following data packets (multiple) in the number equal to the number of tags written (no duplicate data):

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	

2	Len	1		
3	Address	1		
4	Cmd	1	0x42	
5.1	Data	TagCount	2	Length of data written. 16 bits.
5.2		DataLen	1	The EPC length of the tag being operated on (PC +EPC). The unit is bytes.
5.3		EPC	random length	The tag EPC is operated. PC (2 bytes) + EPC (according to tag specification)
5.4		ErrCode	1	The result of the operation on the tag, i.e. error code.
5.5		AntID	1	Antenna number.
6	Checksum	1		

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x 4 2	
5	Data	1	Error code	0x11: Operation failed 0x33: Write tag error 0x58: Two-way authentication failed 0x59: Two-way authentication is successful 0x36: An unoperable tag 0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter 0x43: wordcnt parameter exceeds the specified limit 0x60: Insufficient power supply for the tag 0x61: Insufficient tag permissions 0x62: Memory address is out of range 0x63: Memory is locked 0x64: Incorrect operation password 0x65: Tag authentication reader failed 0x66: Unknown error
6	Checksum	1		

3.4. 43H — — GB-standard encryption communication (read) for Fudan Micro tags

Command: cmd_gb_seu_read (0x43) Before executing this command operation, you must do a two-way authentication!

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	0x0 C	
3	Address		1		
4	Cmd		1	0x 43	
5.1	Data	MemBank Tag storage area	1		0 tag information area 1 Coding area 2 Safe zone 3 User area
5.2		WordAdd Read the first address of the data	2		Refer to the tag specifications for the range of values
5.3		WordCnt Read the data length	2		Word length, WORD (16 bits) length. Refer to the tag specification for the range of values.
5.4		Password	4		Enter the password for the tag
6	Checksum		1		

The reader returns the data packet

Returns the following data packets (multiple) in the number equal to the number of tags read (no duplicate data):

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x43	
5.1	Data	TagCount	2		Total number of successfully operated tags. 16 bits.
5.2		EPCLen	1		The EPC length of the tag being operated on (PC +EPC). The unit is bytes.
5.3		EPC	random length		The EPC of the tag being operated. (PC +EPC).
5.4		ReadLen	2		Read the length of data to be read. The

				unit is bytes.
5.5		ReadData	random length	Read the data being operated on
5.6		AntID	1	Antenna number.
6	Checksum	1		

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x43	
5	Data	1	Error code	0x11: Operation failed 0x32: Tag read error 0x58: Two-way authentication failed 0x59: Two-way authentication is successful 0x36: An unoperable tag 0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter 0x43: wordcnt parameter exceeds the specified limit 0x60: Insufficient power supply for the tag 0x61: Insufficient tag permissions 0x62: The memory address is out of range 0x63: Memory is locked 0x64: Incorrect operation password 0x65: Tag authentication reader failed 0x66: Unknown error
6	Checksum	1		

3.5. 45H—— Read GB-standard tags

Command: cmd_gb_read (0x45)

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
-----	-------	-------------	-------------	---------

1	Head	1	0xA0	
2	Len	1	0x0 C	
3	Address	1		
4	Cmd	1	0x 45	
5.1	Data	MemBank Tag storage area	1	0 tag information area 1 Coding area 2 Safe zone 3 User Area
5.2		WordAdd Read the first address of the data	2	Refer to the tag specifications for the range of values
5.3		WordCnt Read the data length	2	Word length, WORD (16 bits) length. Refer to the tag specification for the range of values.
5.4		Password	4	Enter the password for the tag
6	Checksum	1		

Reference example (for reference only):

Data frame header Head: 0xA0
Packet length Len: 0x0C
Reader address Address: 0x00
command code Cmd: 0x45
data packet Data: 0x 00 0002 0001 00000000

//00 Tag information area
//0002 Read the first address of the data
//0001 Read data length
//00000000 Tag access password

Verification code Checksum: 0x0C

The reader returns the data packet

Returns the following data packets (multiple) in the number equal to the number of tags read (no duplicate data):

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1		
3	Address	1		
4	Cmd	1	0x45	
5.1	Data	TagCount	2	Total number of successfully operated tags. 16 bits.
5.2		EPCLen	1	The EPC length (PC +EPC) of the tag being operated. The unit is bytes.
5.3		EPC	random	The EPC of the tag being operated. (PC

			length		+EPC).
5.4		ReadLen	2		Read the length of data to be read. The unit is bytes.
5.5		ReadData	random length		Read the data being operated on
5.6		AntID	1		Antenna number.
6	Checksum		1		

Reference example (for reference only):

Data frame header Head: 0xA0
Packet length Len: 0x0C
Reader address Address: 0x00
command code Cmd: 0x45
data packet Data: 0x0001 12 3000E280689400005016A9878056 0002 D578

01

// From high to low 0001 is the total number of successful operations 1,

//12 is the valid data for data operation

//3000 is the PC that reads the data,

// E280689400005016A9878056 is the tag EPC

//0002 Read the length of the data to be operated

//D578 read Operate data

//01 is the antenna number

Verification code Checksum: 0x0E

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x81	
5	Data	1	Error code	0x11: Operation failed 0x32: Tag read error 0x36: An unoperable tag 0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter 0x43: wordcnt parameter exceeds the specified limit 0x60: Insufficient power supply for the tag 0x61: Insufficient tag permissions
				0x62: Memory address is out of range 0x63: Memory is locked 0x64: Incorrect operation password 0x65: Tag authentication reader failed 0x66: Unknown error

6	Checksum	1		
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Reference example (for reference only):

Data frame header Head: 0xA0
Packet length Len: 0x04
Reader address Address: 0x00
command code Cmd: 0x45
data packet Data: 0x36
Verification code Checksum: 0xE1

3.6. 46H—— Write GB-standard tags

Command: cmd_gb_write (0x46)

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x46	
5.1	Data	Password	4		Enter the password for the tag
5.2		MemBank Tag storage area	1		0 tag information area 1 Coding area 2 Safe zone 3 User area
5.3		WordAdd Data first address	2		WORD (16-bit) address. Writing into the EPC storage area usually starts from 01, and the first two bytes of this area store the PC.
5.4		WordCnt Write the data length	2		WORD (16 bits) length, refer to the tag specification for the value
5.5		WordData	WordCnt *2		The data written (variable length) is twice the length of the data written.
6	Checksum		1		

The reader returns the data packet

Returns the following data packets (multiple) in the number equal to the number of tags written

(no duplicate data):

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x46	
5.1	Data	TagCount	2		Length of data written. 16 bits.
5.2		DataLen	1		The EPC length of the tag being operated on (PC +EPC). The unit is bytes.
5.3		EPC	random length		The tag to be operated is EPC. PC (2 bytes) + EPC (according to tag specification)
5.4		ErrCode	1		The result of the operation on the tag, i.e., the error code.
5.5		AntID	1		Antenna number.
6	Checksum		1		

Failure return:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	4	
3	Address		1		
4	Cmd		1	0x 46	
5	Data		1	Error code	0x11: Operation failed 0x33: Tag writing error 0x36: An unoperable tag 0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter 0x43: wordcnt parameter exceeds the specified limit 0x60: Insufficient power supply for the tag 0x61: Insufficient tag permissions 0x62: The memory address is out of range 0x63: Memory is locked 0x64: Incorrect operation password 0x65: Tag authentication reader failed 0x66: Unknown error

6	Checksum	1		
---	----------	---	--	--

3.7. 47H—— Lock GB-standard tags

Command: cmd_gb_lock

host computer sends command data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	0x09	
3	Address		1		
4	Cmd		1	0x47	
5.1	Data	Password	4		Enter the password for the tag
5.2		MemBank Tag storage area	1		0x00 tag information area 0x01 coding area 0x02 Security zone 0x03 User area
5.3		LockType Lock operation type	1		0x00, readable and writable 0x01 Read only 0x02 Not readable or writable 0x03 Not readable or writable 0x04 Reserved 0x05 No identification required 0x06 Security communication is not required for authentication 0x07 Security communication needs to be identified
6	Checksum		1		

Reader returns packet:

Returns the following data packets (multiple) in the number equal to the number of tags written (no duplicate data):

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x 47	
5.1	Data	TagCount	2		Total number of successful operation tags. 16 bits.
5.2		DataLen	1		The length of the valid data in the tag being operated on. (PC+CRC+EPC). The

				unit is bytes.
5.3		Lock Data	random length	Operate valid data of the tag. PC (2 bytes) + EPC (according to tag specification)
5.4		ErrCode	1	The result of the operation on the tag, i.e., the error code.
5.5		AntID	1	The high 6 bits are the first read frequency parameters, and the low 2 bits are the antenna number.
6	Checksum		1	

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x 47	
5	Data	1	Error code	0x11: Operation failed 0x32: Tag read error 0x36: An unoperable tag 0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter 0x60: Insufficient power supply for the tag 0x61: Insufficient tag permissions 0x62: The memory address is out of range 0x63: Memory is locked 0x64: Incorrect operation password 0x65: Tag authentication reader failed 0x66: Unknown error
6	Checksum	1		

3.8. 49H—— Kill GB-standard tags

Command: cmd_gb_kill

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	7	
3	Address		1		
4	Cmd		1	0x 49	
5.1	Data	Password	4		tag inactivate password
6	Checksum		1		

The reader returns the data packet

Returns the following data packets (multiple) in the number equal to the number of tags destroyed (no duplicate data):

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x 49	
5.1	Data	TagCount	2		Total number of successfully operated tags. 16 bits.
5.2		DataLen	1		The effective data length of the tag being operated on (PC +EPC). The unit is bytes.
5.3		Kill Data	random length		Operate valid data of the tag. PC (2 bytes) + EPC (according to tag specification)
5.4		ErrCode	1		The result of the operation on the tag, i.e. error code.
5.5		AntID	1		The high 6 bits are the first read frequency parameters, and the low 2 bits are the antenna number.
6	Checksum		1		

Failure return:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	4	
3	Address		1		
4	Cmd		1	0x 49	
5	Data		1	Error code	0x11: Operation failed 0x32: Tag read error 0x36: An unoperable tag
					0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter 0x60: Insufficient power supply for the

6	Checksum	1		
---	----------	---	--	--

3.9. 4AH——Save parameters

Command: cmd_reader_para_save

This command writes the configured parameters into FLASH, ensuring data is not lost during power failure. Among all commands for setting parameters, except for the communication bandwidth command, this command must be used after configuring other parameters to ensure their permanent storage. If multiple parameters need to be configured, send another parameter saving command after completing all parameter configurations, and all configured parameters will be saved in FLASH.

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x4a	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 4A 13

Data analysis: None

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x4a	
5	Data	1	Error code	0x10: Operation successful 0x11: Operation failed
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 4A 10 02

Data analysis: 10: Operation successful

3.10. 4BH—— Restore the default configuration of reader

Command: cmd_reader_para_reset

This command restores all parameters of the reader except system time and MAC address to their default values, including RFID configuration parameters.

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x4b	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 4B 12

Data analysis: None

The reader does not return a data packet

3.11. 4CH—— Application update

Command: cmd_reader_app_upgrade

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x4c	
5.1	Data	PacketSerialNumber	4		Upgrade the data packet serial number to start with 0x00000000 and end with 0xFFFFFFFF
5.2		Status	1		0, during the upgrade; 1, end of the

					upgrade (last package)
5.3		upgradeDataLen	1		N, the length of upgradeData
5.4		upgradeData	random length		Upgrade data content
6	Checksum		1		

Reference example:

Serial port data: None

Data analysis: None

Reader returns packet:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	8	
3	Address		1		
4	Cmd		1	0x4c	
5.1	Data	PacketSerialNumber	4		The sequence number of the upgrade package sent by the host computer
5.2		Status	1		0 Success 1. Failure
6	Checksum		1		

Reference example:

Serial port data: None

Data analysis: None

3.12. 4DH—— Firmware update

Command: cmd_baseband_firmware_upgrade

Upgrade the reader application software

Reference example:

Serial port data: None

Data analysis: None

3.13. 4EH—— Set the heartbeat switch and cycle

Command: cmd_set_keepalive

The host machine sends instruction data

NO.	field	NO.of bytes	fixed value	remarks
-----	-------	-------------	-------------	---------

1	Head	1	0xA0	
2	Len	1	6	
3	Address	1		
4	Cmd	1	0x4E	
5.1	Data	switch	1	0x1 Open the heart packet 0x0 Turn off heartbeat packets
5.2		Heartbeat cycle	2	The minimum value is 100ms, the maximum is 655350ms, the minimum modification unit is 10ms, and the big-endian mode is used
6	Checksum	1		

Reference example:

Serial port data: A0 06 00 4E 01 13 88 70

Data analysis: 01: Enable heartbeat packets

13 88: Heartbeat cycle 5000 (big-endian mode)

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x4E	
5	Data	1	0x10	Operation successful
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 4E 10 FE

Data analysis: 10: Operation successful

3.14. 4FH—— Get the heartbeat switch and cycle

Command: cmd_get_keepalive

The host machine sends instruction data

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x4F	
6	Checksum	1		

Reference example:

Serial port data: A0 03 00 4F 0E

Data analysis: None

Reader returns packet:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	6	
3	Address		1		
4	Cmd		1	0x4F	
5.1	Data	switch	1		0x1 Heartbeat packet has been opened 0x0 Heartbeat packet has been closed
5.2		Heartbeat cycle	2		(Large end mode)
6	Checksum		1		

Reference example:

Serial port data: A0 06 00 4F 00 27 10 D4

Data analysis: 00: Heart rate switch

27 10: Heartbeat cycle 10000 (big-endian mode)

3.15. 5AH—— Get the session and target

Command: cmd_get_session_target

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	3	
3	Address		1		
4	Cmd		1	0x5A	
5	Checksum		1		

Reference example:

串口数据:A003005A 03.

Data analysis: None

Reader returns packet:

NO.	field		NO.of bytes	fixed value	remarks
-----	-------	--	-------------	-------------	---------

1	Head		1	0xA0	
2	Len		1	5	
3	Address		1		
4	Cmd		1	0x5A	
5.1	Data	Session	1		Specify the session for inventory 0 S0 1 S1 2 S2 3 S3
5.2		Target	1		Specify the inventory Inventoried Flag 0 A 1 B
6	Checksum		1		

Reference example:

Serial port data: A0 05 00 5A 02 01 FE

Data analysis: 02: session-s2

01: target-B

3.16. 5BH—— Set the Session and Target

Command: cmd_set_session_target

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	5	
3	Address		1		
4	Cmd		1	0x5B	
5.1	Data	Session	1		Specify the session for inventory 0 S0 1 S1 2 S2 3 S3
5.2		Target	1		Specify the Inventoried Flag for inventory 0 A 1 B
6	Checksum		1		

Reference example:

Serial port data: A0 05 00 5B 02 01 FD

Data analysis: 02: session-s2

01: target-B

Reader returns packet:

Reference example:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x5B	
5	Data	1	0x10	Operation successful
6	Checksum	1		

Serial port data: A0 04 00 5B 10 F1

Data analysis: 10: Operation successful

3.17. 5CH—— Get TX RF duty cycle

Command: cmd_get_tx_time

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x5C	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 5C 01

Data analysis: None

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	7	
3	Address	1		
4	Cmd	1	0x5C	
5.1	Tx_RF	Tx_RF_ontime	2	The step is 1ms
5.2		Tx_RF_offtime	2	The step is 1ms
6	Checksum	1		

Reference example:

Serial port data: A0 07 00 5C 00 64 00 64 35

Data analysis: 00 64: Tx_RF_ontime (big-endian mode)

00 64: Tx_RF_offtime (big-endian mode)

3.18. 5DH—— Set TX RF time

Command: cmd_set_tx_time

This command is mainly used to change the TX RF duty cycle and adjust the power consumption. The TX transmission time is mainly divided into TX RF enable time and TX RF off time. The minimum time scale is 10ms, and the step is 1ms.

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	7	
3	Address		1		
4	Cmd		1	0x5D	
5.1	Tx_RF	Tx_RF_ontime	2		The default value is 100ms, and the step is 1ms. Both fields can be set to 0 at the same time to indicate that the function is turned off
5.2		Tx_RF_offtime	2		
6	Checksum		1		

Reference example:

Serial port data: A0 07 00 5D 00 64 00 64 34

Data analysis: 00 64: Tx_RF_ontime (big-endian mode)

00 64: Tx_RF_offtime (big-endian mode)

Reader returns packet:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	4	
3	Address		1		
4	Cmd		1	0x5D	
5	Data		1		0x10: The setting is successful
6	Checksum		1		

Reference example:

Serial port data: A0 04 00 5D 10 EF

Data analysis: 10: Operation successful

3.19. 5FH—— Set the RF output power for 4-antenna reader

Command: cmd_set_4_ant_power

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	7	
3	Address	1		
4	Cmd	1	0x5F	
5	Data	4		Byte 1: RF output power of antenna 1 The second byte: RF output power of antenna 2 The third byte: RF output power of antenna 3 Byte 4: RF output power of antenna 4 The range is 0–33, and the unit is dBm
6	Checksum	1		

Note: The new reader configuration takes effect immediately.

Reference example:

Serial port data: A0 07 00 5F 0A 21 21 21 8D

Data analyze:

0A: RF output power of antenna 1 (10dBm)

21: RF output power of antenna 2 (33dBm)

21: RF output power of antenna 3 (33dBm)

21: RF output power of antenna 4 (33dBm)

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x5F	
5	Data	1	Error code	0x10: Operation successful 0x11: Operation failed 0x25: Parameter out of range
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 5F 10 ED

Data analysis: 10: Operation successful

3.20. 60H—— Get the GPIO status

Command: cmd_read_gpio_value

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x60	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 60 FD

Data analysis: None

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	5	
3	Address	1		
4	Cmd	1	0x60	
5	Data	1	GPIO1 state	0: Low
		1	GPIO2 state	1: High
6	Checksum	1		

Reference example:

Serial port data: A0 05 00 60 00 00 FB

Data analysis: 00: GPIO1 status-low

00: GPIO2 status-low

3.21. 61H—— Set the GPIO status

Command: cmd_set_gpio_value

host computer sends instruction data:

NO.	field	NO.of	fixed value	remarks
-----	-------	-------	-------------	---------

		bytes		
1	Head	1	0xA0	
2	Len	1	5	
3	Address	1		
4	Cmd	1	0x61	
5	Data	1	GPIO port selection	0x03:GPIO0 0x04:GPIO1
		1	GPIO output value	0: Low 1: High
6	Checksum	1		

Reference example:

Serial port data: A0 05 00 61 03 01 F6

Data analysis: 03: GPIO0

01: Set it to high

Reference example:

Serial port data: A0 05 00 61 04 01 F5

Data analysis: 04: GPIO1

01: Set it to high

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x61	
5	Data	1	Error code	0x10: Operation successful 0x11: Operation failed
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 61 10 EB

Data analysis: 10: Operation successful

3.22. 66H — — Set the temporary RF output power of the reader

Note: 4 This command is a debugging command. To set the antenna power, please use the 0x5F command

Command: cmd_set_temporary_output_power

After the operation is successful, the output power value will not be saved in

the internal Flash, and the output power will be restored to the output power value saved in the internal Flash after reboot or power failure. This command operates very quickly and does not write to the Flash, thus not affecting the service life of the Flash, which is suitable for applications that need to switch the RF output power repeatedly.

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x66	
5	Data	1		The RF temporary output power ranges from 20 to 33, in dBm
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 66 11 E5

Data analysis: 11: RF temporary output power

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x66	
5	Data	1	Error code	0x10: Operation successful 0x11: Operation failed 0x25: Parameter out of range
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 66 10 EB

Data analysis: 10: Operation successful

3.23. 69H—— Configure the RF link parameters

Command: cmd_set_rf_link_profile

After the operation is successful, the reader will restart and the configuration will be directly saved in the internal Flash without loss after power failure. If other parameters are configured before configuring this parameter, the previously configured parameters will also be written into the Flash.

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x69	
5	Data	1		ProfileID 0xD0 Miller er 40KHz tari 25us 0xD1 FM0 200KHz tari 6.25us 0xD2 FM0 200KHz tari 12.5us 0xD3 FM0 200KHz tari 25us 0xD4 Miller4 200KHz tari 6.25us 0xD5 Miller4 200KHz tari 12.5us 0xD6 Miller4 200KHz tari 25us 0xD7 Miller_4 250KHz tari 6.25us 0xD8 FM0 640KHz tari 6.25us 0xD9 FM0 40KHz tari 25us 0xDA GB FM0 64KHz 0xDB GB Miller 128KHz 0xDC GB FM0 128KHz
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 69 D7 1C

Data analysis: D7: Miller_4 250KHz tari 6.25us

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x69	
5	Data	1	Error code	0x10: Operation successful 0x11: Operation failed
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 69 10 E3

Data analysis: 10: Operation successful

3.24. 6AH—— Get the RF link parameters

Command: cmd_get_rf_link_profile

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x6a	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 6A F3

Data analysis: None

The reader returns the data packet

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x6a	
5	Data	1		Success returned, ProfileID 0xD0 Mill er 40KHz tari 25us 0xD1 FM0 200KHz tari 6.25us 0xD2 FM0 200KHz tari 12.5us 0xD3 FM0 200KHz tari 25us 0xD4 Miller4 200KHz tari 6.25us 0xD5 Miller4 200KHz tari 12.5us 0xD6 Miller4 200KHz tari 25us 0xD7 Miller_4 250KHz tari 6.25us 0xD8 FM0 640KHz tari 6.25us 0xD9 FM0 40KHz tari 25us 0xDA GB FM0 64KHz 0xDB GB Miller 128KHz 0xDC GB FM0 128KHz
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 6A D6 1C

Data analysis: D6: Miller4 200KHz tari 25us

3.25. 70H — Reset reader

Command: cmd_reset

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x70	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 70 ED

Data analysis: None

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x70	
5	Data	1	Error code	0x10: Operation successful 0x11: Operation failed 0x20: MCU reset failure
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 70 10 DC

Data analysis: 10: Operation successful

3.26. 71H—— Configure the serial communication baud rate

Command: cmd_set_uart_baudrate

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x71	
5	Data	1		0x00:1200 0x01:2400 0x02:4800 0x03:9600 0x04:14400 0x05:19200

				0x06:38400 0x07:57600 0x08:76800 0x09:115200 0x0a:128000 0x0b:230400 0x0c:256000 0x0d:460800
6	Checksum	1		

Note: The command frame is successfully received by the reader, and the command takes effect immediately, updating the reader baud rate.

Reference example:

Serial data: A0 04 00 71 09 E2

Data analysis: 09:115200

The reader returns the data packet

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x71	
5	Data	1	action code	0x10: Operation successful 0x11: Operation failed 0x4a: The baud rate is out of range
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 71 10 DB

Data analysis: 10: Operation successful

3.27. 72H—— Get the firmware version

Command: cmd_get_firmware_version

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x72	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 72 EB

Data analysis: None

The reader returns the data packet

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	6	
3	Address		1		
4	Cmd		1	0x72	
5.1	Data	Major	1		The main version number of the firmware
5.2		Minor	1		Firmware subversion number
5.3		Model	1		0x01:MU601 0x02:MU602 0x06:MU606 0x08:MU608
6	Checksum		1		

Reference example:

Serial port data: A0 06 00 72 02 04 01 E1

Data analysis: 02: Main version number

04: Subversion number

01: MU601

3.28. 73H—— Configure the RS-485 network port address

Command: cmd_set_reader_address

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	4	
3	Address		1		
4	Cmd		1	0x73	
5	Data		1		Configure the reader address, configuration range 0-255
6	Checksum		1		

Note: The new reader address takes effect immediately.

Reference example:

Serial port data: A0 04 00 71 10 DB

Data analysis: 10: Operation successful

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x73	
5	Data	1	Error code	0x10: Operation successful 0x41: The configuration parameter is not supported
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 73 10 D9

Data analysis: 10: Operation successful

3.29. 74H—— Set the current working antenna port

Command: cmd_set_work_antenna

host computer sends instruction data--single antenna:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x74	
5	Data	1		Antenna number AntennaID Antenna 1, 0x01
6	Checksum	1		

Note: The new reader configuration takes effect immediately.

Reference example:

Serial port data: A0 04 00 74 01 E7

Data analysis: 01: Enable antenna 1

host computer sends instruction data--multiple antennas (only MU608-4 and MU608-8 are valid):

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x74	
5	Data	1	0x00	fixed value

		8		1~8 bytes correspond to 1~8 antenna enable bits. If the antenna is opened, the corresponding byte is set to 1; if the antenna is closed, the corresponding byte is set to 0. The first 4 bytes of the 4-antenna module are valid
6	Checksum	1		

Reference example:

Serial port data: A0 0C 00 74 00 01 01 00 00 00 00 00 DE

Data analysis: 00: Fixed value

Enable antenna 1

Enable antenna 2

00: Except for antenna 3

00: Except for the 4th antenna

00: Except for the 5 antenna

00: Except for the 6th antenna

00: Except for antenna 7

00: Except for the 8th antenna

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x74	
5	Data	1	Error code	0x10: The operation was successful 0x41: The configuration parameter is not supported
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 74 10 D8

Data analysis: 10: Operation successful

3.30. 75H—— Get the current working antenna port

Command: cmd_get_work_antenna

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	

3	Address	1		
4	Cmd	1	0x75	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 75 E8

Data analysis: None

The reader returns the data packet

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x75	
5.1	Data	1		Antenna number AntennaID Antenna 1,0x01 Antenna 2, 0x02 Antenna 3,0x03 Antenna 4, 0x04
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 75 01 E6

Data analysis: 01: Antenna 1 is working

3.31. 76H — — Set the RF output power of single antenna reader

Command: cmd_set_output_power

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x76	
5	Data	1		RF output power, ranging from 0 to 33, in dBm (601 series supports 20dbm, 606L 26dbm, 608 33dbm, four antennas, eight antennas are set the same)
6	Checksum	1		

Note: The new reader configuration takes effect immediately.

Reference example:

Serial port data: A0 04 00 76 0A DC

Data analysis: 0A: RF output power (10dBm)

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x76	
5	Data	1	Error code	0x10: Operation successful 0x11: Operation failed 0x25: Parameter out of range
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 76 10 D6

Data analysis: 10: Operation successful

3.32. 77H—— Get the current RF output power

Command: cmd_get_output_power

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x77	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 77 E6

Data analysis: None

Reader returns data packet:

Situation 1: Power of a single antenna

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	

3	Address	1		
4	Cmd	1	0x77	
5	Data	1		Antenna 1 power, range 0-33, unit dBm
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 77 14 D1

Data analysis: 14: Antenna 1 power

Situation 2: Power of 4 antennas

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	7	
3	Address		1		
4	Cmd		1	0x77	
5.1	Data	Power1	1		Antenna 1 power, range 0-33, unit dBm
5.2		Power2	1		Antenna 2 power, range 0-33, unit dBm
5.3		Power3	1		Antenna 3 power, range 0-33, unit dBm
5.4		Power4	1		Antenna 4 power, range 0-33, unit dBm
6	Checksum		1		

Reference example:

Serial port data: A0 07 00 77 0A 21 21 21 75

Data analysis: 0A: Antenna 1 power

21: Antenna 2 power

21: Antenna 3 power

21: Antenna 4 power

3.33. 78H—— Set the operating frequency range

Command: cmd_set_frequency_region

host computer sends instruction data:

There are two ways to define the radio frequency spectrum.

Method 1: Use the default frequency point of the system (see the corresponding table of frequency parameters)

CHN-1 is 840-845MHz

The CHN-2 is 920-925Mhz

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	6	
3	Address		1		

4	Cmd		1	0x78	
5.1	Data	Region Radio frequency specifications	1		0x01 FCC 0x02 ETSI 0x03 CHN_1 0x05 CHN_2 When the system default frequency is used (RF specifications 1,2,3,5), the frequency starting point and ending point are configured by default
5.2		StartFreq Frequency starting point	1		You can set the frequency range of frequency hopping within the frequency range specified by the RF specification. Please refer to the frequency parameter correspondence table for the corresponding frequency of the parameters. The setting rules of the parameters are as follows: 1. The starting frequency and ending frequency cannot exceed the range of the RF specification. 2. The starting frequency must be lower than the ending frequency. 3. If the starting frequency is equal to the ending frequency, the fixed frequency transmission is determined.
5.3		EndFreq Frequency end point	1		
6	Checksum		1		

Reference example:

Serial port data: A0 06 00 78 01 07 3B 9F

Data analysis: 01: RF specification FCC

07: FCC Standard frequency starting point (902.250MHz)

3B: FCC Standard frequency end point (928.250MHz)

Method 2: User-defined spectrum

Note: When using a custom spectrum (RF specification 4), set the frequency range according to the starting and ending points of the frequency

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	10	
3	Address	1		
4	Cmd	1	0x78	

5.1	Data	Region Radio frequency specifications	1	4	fixed value
5.2		FreqSpace Frequency spacing	2		Frequency interval = The minimum frequency interval is 25KHz
5.3		FreqQuantity Frequency count	1		The number of frequency points including the starting frequency. 1 means that the frequency is set at the starting frequency. This parameter must be greater than 0
5.4		StartFreq Frequency starting point	3		The unit is KHz. The high order of the hexadecimal number is in front. For example, 915000KHz is sent as 0D F6 38
6	Checksum		1		

Reference examples:

Serial port data: A0 0A 00 78 04 00 19 01 0D BB A0 58

Data analysis: 04: RF specification-custom

019: Frequency band interval (25KHz)

01: Number of frequency points

0D BB A0: Starting frequency (900000KHz)

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x78	
5	Data	1	Error code	0x10: Operation successful 0x11: Operation failed
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 78 10 D4

Data analysis: 10: Operation successful

3.34. 79H—— Get the operating frequency range

Command: cmd_get_frequency_region

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
-----	-------	-------------	-------------	---------

1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x79	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 79 E4

Data analysis: None

According to the RF specification, the data returned by the reader is mainly divided into two categories:

The system returns data at the default frequency

CHN-1 is 840-845MHz

The CHN-2 is 920-925Mhz

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	6	
3	Address		1		
4	Cmd		1	0x79	
5.1	Data	Region Radio frequency specifications	1		0 x01 FCC 0x02 ETSI 0x03 CHN_1 0x05 CHN_2
5.2		StartFreq Frequency starting point	1		The low point of the frequency range for hopping
5.3		EndFreq Frequency end point	1		The high point of the frequency range of the hop
6	Checksum		1		

Reference example:

Serial port data: A0 06 00 79 05 2B 35 7C

Data analysis: 05: RF specification CHN_2

07: CHN_2 standard frequency starting point (920MHz)

3B: CHN_2 standard frequency end point (925MHz)

Return data from a custom frequency

NO.	field		NO.of bytes	fixed value	remarks
-----	-------	--	-------------	-------------	---------

1	Head		1	0xA0	
2	Len		1	10	
3	Address		1		
4	Cmd		1	0x79	
5.1	Data	Region Radio frequency specifications	1	4	fixed value
5.2		FreqSpace Frequency spacing	2		Frequency interval = The minimum frequency interval is 25KHz
5.3		FreqQuantity Frequency count	1		The number of frequency points including the starting frequency. 1 means that the frequency is set at the starting frequency. This parameter must be greater than 0
5.4		StartFreq Frequency starting point	3		The unit is KHz. The high order of the hexadecimal number is in front. For example, 915000KHz is sent 0D F6 38
6	Checksum		1		

Reference example:

Serial port data: A0 0A 00 79 04 00 19 01 0D BB A0 57

Data analysis: 04: RF specification-custom

00 19: Frequency band interval (25KHz)

01: Number of frequency points

0D BB A0: Starting frequency (900000KHz)

3.35. 7AH—— Configure the buzzer

Command: cmd_set_buzzer_switch

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x7a	
5	Data	1		0: Turn off the buzzer 1: Turn on the buzzer
5	Checksum	1		

Reference example:

Serial port data: A0 04 00 7A 00 E2

Data analysis: 00: Turn off the buzzer

Reader returns packet:

Successful return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x7a	
5.1	Data	1	Error code	10: Operation successful 11: Operation failed
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 7A 10 D 2

Data analysis: 10: Operation successful

3.36. 7BH—— Get the operating temperature

Command: cmd_get_reader_temperature

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x7b	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 7B E2

Data analysis: None

Reader returns packet:

Successful return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	5	
3	Address	1		
4	Cmd	1	0x7b	
5.1	Data	PlusMinus	1	Zero below 1 above zero
5.2		Temperature	1	degree Celsius

6	Checksum	1		
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Reference example:

Serial port data: A0 05 00 7B 01 1D C2

Data analysis: 01: above zero

1D: temperature 29°

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x7b	
5	Data	1	Error code	0x11: Operation failed
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 7B 11 D0

Data analysis: 11: Operation failed

3.37. 80H—— Inventory Tags

Command: cmd_inventory

Cache mode: After receiving this command, the reader performs multi-tag identification operation. The tag data is stored in the reader cache area, and the tag data can be obtained by using the extract cache command. See: Cache operation command. Before receiving the stop command, the reader will continue to store inventory.

host computer sends command data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x80	
5	Data	1		Single antenna product: The antenna number is fixed to 0x01 Multi-antenna products: Single antenna mode: antenna number 0x01~0x04 Multi-antenna mode: The antenna number is fixed at 0x00

6	Checksum	1		
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Reference example:

Serial port data: A0 04 00 80 02 DA

Data analysis: 02: Antenna 2

The reader returns the data packet

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x80	
5	Data	1	Error code	0x10: Operation successful 0x11: Operation failed 0x36: An unoperable tag
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 80 10 CC

Data analysis: 10: Operation successful

When the operation tag is not available, continue to return:

Serial port data: A0 04 00 80 36 A6

Data analysis: 36: Inoperable tag

When an actionable tag is available, continue back:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	5	
3	Address	1		
4	Cmd	1	0x80	
5	Data	2		The total number of tags to be identified is distinguished according to the EPC number, and tags with the same EPC number will be regarded as one tag. If the cache is not cleared, the number of tags is the sum of multiple inventory operations.
6	Checksum	1		

Reference example:

Serial port data: A0 05 00 80 00 02 D9

Data analysis: 00 02: Total number of identified tags

3.38. 81H—— Read ISO/IEC 18000-6C tags

Command: cmd_read

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	0x0E	
3	Address		1		
4	Cmd		1	0x81	
5.1	Data	MemBank Tag storage area	1		0 RESERVED 1 EPC 2 TID 3 USER
5.2		WordAdd Read the first address of the data	4		Refer to the tag specifications for the range of values
5.3		WordCnt Read the data length	2		Word length, WORD (16 bits) length. Refer to the tag specification for the range of values.
5.4		Password	4		Enter the password for the tag
6	Checksum		1		

Reference example:

Serial port data: A0 0E 00 81 01 00 00 00 02 00 06 00 00 00 00 C8

Data analysis: 01: Tag storage area EPC

00 00 00 02: Data first address

00 06: Read data length (6 words, 12 bytes)

00 00 00 00: Access password

The reader returns the data packet

Return the following data packet, one piece of data at a time:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x81	
5.1	Data	TagCount	2		Total number of successfully operated

				tags (big-endian mode)
5.2	DataLen	1		Effective data length (PC+CRC+EPC+read data). The unit is bytes.
5.3	Read Data	random length		Valid data for the tag being operated. PC (2 bytes) + EPC (according to tag specification) + CRC (2 bytes) + read data. (If the epc starts from 0, the order is crc+pc+epc) //(PC (2 bytes) + EPC + CRC (2 bytes) is the entire content of the EPC storage area.)
5.4	ReadLen	2		Read the length of data read. The unit is bytes.
5.5	AntID	1		The high 6 bits are the first read frequency parameters, and the low 2 bits are the antenna number.
5.6	ReadCount	1		The number of times the tag has been successfully operated.
6	Checksum	1		

Reference example:

Serial port data: A0 26 00 81 00 01 1C 34 00 E2 00 00 02 02 03 04 05 06 07 08 09 E1 9F E2 00 00 02 02 03 04 05 06 07 08 09 00 0C 03 01 B8

Data analysis: 00 01: Total number of successful operation tags

1C: Effective data length (PC+EPC+CRC+read data)

34 00: PC

E2 00 00 02 02 03 04 05 06 07 08 09: EPC

E1 9F: CRC

E20000020203040506070809: Data read

00C: The length of the data read is 12 bytes

03: Antenna number

01: Number of successful reads

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x81	
5	Data	1	Error code	0x11: Operation failed 0x32: Tag read error 0x36: An unoperable tag

				0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter 0x43: wordcnt parameter exceeds the specified limit
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 81 00 40 9B

Data analysis: 40: Access tag error or password error

3.39. 82H——Write ISO/IEC 18000-6C tags

Command: cmd_write (0x82)

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x82	
5.1	Data	Password	4		Enter the password for the tag
5.2		MemBank Tag storage area	1		0 RESERVED 1 EPC 2 TID 3 USER
5.3		WordAdd Data first address	4		WORD (16 bits) address. Writing into the EPC storage area generally starts from 02, and the first four bytes of this area store PC+CRC.
5.4		WordCnt Write the length of data	2		WORD (16 bits) length, refer to the tag specification for the value
5.5		WordData	WordCnt *2		The data written (variable length) is twice the length of the data written.
6	Checksum		1		

Reference example:

Serial port data: A0 10 00 82 00 00 00 00 01 00 00 00 02 00 01 00 67 63

Data analysis: 01: Tag storage area EPC

00 00 00 02: Data first address

00 01: Write length (1 word, 2 bytes)

00 67: Data written

The reader returns the data packet

Returns the following data packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x82	
5	Data	1	Error code	0x10: Operation successful
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 82 10 CA

Data analysis: 10: Operation successful

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1		
3	Address	1		
4	Cmd	1	0x82	
5.1	Data	TagCount	2	continue to have
5.2		DataLen	1	The length of the valid data of the tag being operated. (PC+EPC+CRC). The unit is bytes.
5.3		Write Data	random length	Operate valid data of the tag. PC (2 bytes) + EPC (according to tag specification) + CRC (2 bytes) (PC (2 bytes) + EPC + CRC (2 bytes) is the entire content of the EPC storage area.)
5.4		ErrCode	1	The result of the operation on the tag, i.e. the error code.
5.5		AntID	1	Antenna number.
6	Checksum	1		

Reference example:

Serial port data: A0 18 00 82 00 00 10 30 00 E2 80 69 95 00 00 50 05 CF 23 28 62 46 81 10 01 7D

Data analysis: 00 00: reserved

10: Effective data length (PC+EPC+CRC)

30 00: PC

E2 80 69 95 00 00 50 05 CF 23 28 62: EPC

46 81: CRC

10: Operation result: success

01: Antenna number

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x82	
5	Data	1	Error code	0x11: Operation failed 0x33: Write tag error 0x36: An unoperable tag 0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter 0x43: wordcnt parameter exceeds the specified limit
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 82 41 99

Data analysis: 41: Invalid parameter

3.40. 83H—— Lock ISO/IEC 18000-6C tags

Command: cmd_lock

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	9	
3	Address	1		
4	Cmd	1	0x83	
5.1	Data	Password	4	Enter the password for the tag
5.2		MemBank Tag storage area	1	0x01 User Memory 0x02 TID Memory 0x03 EPC Memory 0x04 Access Password 0x05 Kill Password
5.3		LockType Lock operation type	1	0x00 Open 0x01 Lock (precondition for tag lock: the access password is not all 0, and
				the access password needs to be modified in advance. Note: NXP U9 The tag has no access password, so the lock operation cannot be performed) 0x02, permanently open 0x03 permanently locked

6	Checksum	1		
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Reference example:

Serial port data: A0 09 00 83 00 00 00 01 00 D3

Data analysis: 00 00 00 00: tag access password

01: tag storage area user memory

00: Lock operation type, open

Reader returns packet:

Returns the following data packets (multiple) in the number equal to the number of tags written (no duplicate data):

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x83	
5.1	Data	TagCount	2		continue to have
5.2		DataLen	1		The length of the valid data for the tag being operated on. (PC+CRC+EPC). The unit is bytes.
5.3		Lock Data	random length		Operate valid data of the tag. PC (2 bytes) + EPC (according to tag specification) + CRC (2 bytes) (PC (2 bytes) + EPC + CRC (2 bytes) is all the content in the EPC storage area.)
5.4		ErrCode	1		The result of the operation on the tag, i.e., the error code.
5.5		AntID	1		The high 6 bits are the first read frequency parameters, and the low 2 bits are the antenna number.
5.6		Lock Count	1		The number of times the tag has been operated
6	Checksum		1		

Reference example:

Serial port data: A0 18 00 83 00 00 10 34 00 E2 00 00 02 02 03 04 05 06 07 08 09 E1 9F 10 01 E0

Data analysis: 00 00: reserved

10: Effective data length (PC+EPC+CRC)

34 00: PC
E2 00 00 02 02 03 04 05 06 07 08 09: EPC
E1 9F: CRC
10: Operation successful
01: The number of times the tag has been operated

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x83	
5	Data	1	Error code	0x11: Operation failed 0x32: Tag read error 0x36: An unoperable tag 0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 83 36 A3

Data analysis: 36: Inoperable tag

3.41. 84H—— Kill ISO/IEC 18000-6C tags

Command: cmd_kill

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	7	
3	Address	1		
4	Cmd	1	0x84	
5.1	Data Password	4		tag inactivate the password
6	Checksum	1		

Reference example:

Serial port data: A0 07 00 84 00 00 00 01 A3

Data analysis: 00 00 00 01: tag inactivation password

The reader returns the data packet

Returns the following data packets (multiple) in the number equal to the number of tags destroyed (no duplicate data):

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1		
3	Address	1		
4	Cmd	1	0x84	
5.1	Data	TagCount	2	continue to have
5.2		DataLen	1	The length of the valid data for the tag being operated on. (PC+CRC+EPC). The unit is bytes.
5.3		Kill Data	random length	Operate valid data of the tag. PC (2 bytes) + EPC (according to tag specification) + CRC (2 bytes) (PC (2 bytes) + EPC + CRC (2 bytes) is the entire content of the EPC storage area.)
5.4		ErrCode	1	The result of the operation on the tag, i.e. error code.
5.5		AntID	1	The high 6 bits are the first read frequency parameters, and the low 2 bits are the antenna number.
5.6		Kill Count	1	The number of destruction tag operations can only be 1
6	Checksum	1		

Reference example:

Serial port data: A0 19 00 84 00 00 10 30 00 E2 00 00 00 00 00 04 16 A9 87 50 56 22 8E 10 01 01

Data analysis: 00 00: reserved

10: Effective data length (PC+EPC+CRC)

30 00: PC

E2 00 00 00 00 00 04 16 A9 87 50 56: EPC

22 8E: CRC

10: Operation successful

01: Antenna number

01: Number of times the tag is destroyed

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		

4	Cmd	1	0x84	
5	Data	1	Error code	0x11: Operation failed 0x32: Tag read error 0x36: An unoperable tag 0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 84 36 A2

Data analysis: 36: Inoperable tag

3.42. 87H—— Inventory with Multi-antenna polling

Command: cmd_fast_switch_ant_inventory

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1		
3	Address	1		
4	Cmd	1	0x8 7	
5.1	Data	Ant num	1	Number of antennas polled
5.2		Ant1	1	First, the antenna to be polled is set to 0 to skip.
5.3		Stay 1	1	Number of times the antenna repeats polling. Each antenna can be configured separately.
5.4		Ant2	1	The antenna for the second poll is set to 0 to skip.
5.5		Stay 2	1	Number of times the antenna repeats polling. Each antenna can be configured separately.
5.6		Antn	1	The antenna of the n-th polling is set to 0 to skip it.
5.7		Stay n	1	Number of times the antenna repeats polling. Each antenna can be configured separately.
5.8		Interval	1	Rest time between antennas. The unit is mS.

					There is no RF output during rest, which can reduce power consumption.
5.9		Session	1		The designated inventory session, 00 is S0, 01 is S1, 02 is S2, and 03 is S3.
5.10		Flag	1		The designated inventory Inventoried Flag, 00 is A and 01 is B
5.11		Repeat	1		Repeat the above antenna switching sequence times. The maximum is 0xFF
6	Checksum		1		

Reference example:

Serial port data: A0 10 00 87 04 01 01 02 02 03 03 04 04 0A 00 00 01 A6

Data analysis: 04: Poll the number of antennas

01: The antenna number of the first poll

01: The number of times the antenna repeats polling

02: The antenna number of the second poll

02: The number of times the antenna repeats polling

03: The antenna number of the third poll

03: The number of times the antenna repeats polling

04: The first polled antenna number

04: The first polled antenna number

0A: Antenna interval time is 10ms

00: session s0

00: flag A

01: Number of repeats

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x87	
5	Data	1	Error code	0x10: Operation successful
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 87 10 C5

Data analysis: 10: Operation successful

If there is a tag response, return the following packets (multiple):

NO.	field	NO.of bytes	fixed value	remarks
-----	-------	-------------	-------------	---------

1	Head	1	0xA0	
2	Len	1		
3	Address	1		
4	Cmd	1	0x87	
5.1	Data	Ant	1	The antenna number of the current work
5.2		PC	2	The tag PC is fixed with two bytes.
5.3		EPC	random length	The EPC number of the tag can vary in length.
5.4		RSSI	4	Real-time RSSI for the tag
5.5		Freq	3	Read the frequency parameters of the tag
6	Checksum	1		

Reference examples:

Serial port data: A0 19 00 87 01 30 00 E2 80 69 95 00 00 40 02 D7 43 70 AA E8 2B 1C 11 0D F6 38 3E

Data analysis: 01: The antenna number of the current work

30 00: PC

E2 80 69 95 00 00 40 02 D7 43 70 AA: EPC

E8 2B 1C 11: rssi

0D F638: Frequency of the tag to be read

01: Number of repeats

Automatic closing of inventory:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x87	
5	Data	1		0x13: The inventory cycle reaches the end and automatically ends
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 87 13 C2

Data analysis: 13: Number of rounds for inventory arrival

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		

4	Cmd	1	0x87	
5	Data	1	Error code	0x11: Operation failed 0x22: The antenna is not connected 0x36: An unoperable tag
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 87 22 B4

Data analysis: 22: The antenna is not connected

3.43. 89H—— Real-time Inventory

Command: cmd_real_time_inventory

Real-time mode (Auto): After receiving this command, the reader performs multi-tag identification operation. Tag data is uploaded in real time and is not stored in the reader cache. This command takes a long time for disk storage, which is suitable for large batch tag reading. The reader will continue disk storage until it receives a stop command.

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x89	
5	Data	1		Single antenna product: The antenna number is fixed to 0x01 Multi-antenna products: Single antenna mode: antenna number (0x01~0x04) Multi-antenna mode: The antenna number is fixed at 0x00
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 89 01 D2

Data analysis: Antenna 01:1

The reader returns the data packet

If there is a tag response, return the following packets (multiple):

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1		

3	Address	1		
4	Cmd	1	0x89	
5.1	Data	Ant	1	Antenna number
5.2		PC	2	The PC with the tag
5.3		EPC	random length	The length of the EPC number in the tag is variable
5.4		RSSI	4	Real-time RSSI of the tag
5.5		Freq	3	Read back the frequency of the tag
6	Checksum	1		

Reference example:

Serial port data: A0 19 00 89 01 34 00 E2 80 F3 36 20 00 F0 00 0B B7 7B 8D E8 14 30 47 0D F6 38 76

Data analysis: 01: The antenna number of the current work

34 00: PC

E2 80 F3 36 20 00 F0 00 0B B7 7B 8D: EPC

E8 14 30 47: rssi

0D F638: Frequency of the tag to be read

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x89	
5	Data	1	Error code	0x10: Operation successful 0x11: Operation failed 0x22: The antenna is not connected 0x36: An unoperable tag
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 89 10 C3

Data analysis: 10: Operation successful

3.44. 8AH—— Custom Inventory

Command: cmd_custom_inventory

Custom Read Mode: After receiving this command, the reader performs multi-tag identification and custom data reading operation. Tag data is uploaded in real time and is not stored in the reader cache area. This command can customize the number of storage rounds and can customize the tag content to be read.

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	7	
3	Address	1		
4	Cmd	1	0x8 A	
5	Inventory of antennas	1		Single antenna product: The antenna number is fixed to 0x01 Multi-antenna products: Single antenna mode: antenna number (0x01~0x04) Multi-antenna mode: antenna number is fixed at 0x00
6	Inventory count	1		Select the number of rounds to be stored by the reader, 0xFF means always store
7	Match parameters	1		Select the membank that matches the reader 0x00: No match 0x01: Match TID 0x02: Match EPC 0x03: Match USERDATA 0x04: Read the Magnus temperature tag 0x05: National standard matching tag information area 0x06: National standard matching code area 0x0a: Match Kelway tag
8	Match length	1		Indicates the length of data matching membank
9	Checksum	1		

Reference example:

Serial port data: A0 07 00 8A 00 01 02 06 C6

Data analysis: 00: Multi-antenna mode of multi-antenna products

01: Inventory frequency

02: Match EPC

06: Match length

The reader returns the data packet

If there is a tag response, return the following data packet (multiple):

NO.	field	NO.of	fixed	remarks
-----	-------	-------	-------	---------

		bytes	value	
1	Head	1	0xA0	
2	Len	1		
3	Address	1		
4	Cmd	1		Returns the named header based on the selected matching parameters: Mismatches: 0x8A Match TID: 0x53 Match EPC: 0x54 Match USERDATA: 0x55 Read the Magnus temperature tag: 0x52
5.1	Data	Ant	1	The current inventory of antenna numbers
		PC	2	The PC with the tag
		EPC		The EPC number of the tag is variable in length
		CRC	2	The CRC of the tag
		Match data length	1	Return the data length based on the selected matching parameters: Mismatches: 0 Match TID: epc_len Match EPC: epc_len Match USERDATA: 4 Read the Magnus temperature tag: 4
5.2		Data	random length	Returns data based on the length of the matching data
5.3		RSSI	4	Real-time RSSI of the tag
5.4		Freq	3	Read back the frequency of the tag
6	Checksum	1		

Reference example:

Serial port data: A0 28 00 54 01 34 00 E2 80 F3 36 20 00 F0 00 0B B7 31 D9 A1 AD 0C E2 80 F3 36 20 00 F0 00 0B B7 31 D9 E8 09 AA 85 0D EE 68 04

Data analysis: 00: Address

54: Match EPC

01: The current inventory of antenna numbers

34 00: PC

E280 F3362000 F0000B B731 D9: EPC of the tag

A1 AD: CRC

0C: Match data length 12

E280 F3362000 F0000B B731 D9: Match data

E8 09 AA 85: rssi

0D EE 68: Frequency of the tag to be read

Automatic closing of inventory:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1		Select member k from the reader to return the name header: Mismatches: 0x8A Match TID: 0x53 Match EPC: 0x54 Match USERDATA: 0x55 Read the Magnus temperature tag: 0x52
5	Data	1		0x12: Set the number of rounds to reach and automatically exit the inventory
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 54 12 F6

Data analysis: 54: Match EPC

12: Set the number of rounds to reach, and automatically exit the inventory

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x8A	
5	Data	1	Error code	0x11: Operation failed 0x22: The antenna is not connected 0x32: Tag read error 0x36: An unoperable tag 0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter 0x43: wordcnt parameter exceeds the specified limit
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 8A 10 C2

Data analysis: 10: Operation successful

3.45. 8BH — — Configure the session and target for custom inventory

Command: cmd_customized_session_target_inventory

Recommended inventory instructions

Real-time Mode (Session): After receiving this command, the reader performs multi-tag recognition operations according to the specified session and inventoried flag. Tag data is uploaded in real-time and not stored in the readers buffer. The S1 mode is recommended for regular inventory. The reader will continue to perform inventory until it receives a stop command.

For S0~S1 mode, see EPC RFID Protocols_Class1_Gen2_V1.1.0->6.3.2.2 Sessions and inventoried flags for details

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	5	
3	Address		1		
4	Cmd		1	0x8b	
5.1	Data	Session	1		Specify the session for inventory 0 S0 1 S1 2 S2 3 S3
5.2		Target	1		Specify the Inventoried Flag for inventory 0 A 1 B
5.3		Antid	1		Inventory of antennas
6		Checksum	1		

Reference example (for reference only):

Data frame header Head:	0xA0
Packet length Len:	0x06
Reader address Address:	0x00
command code Cmd:	0x8B
data packet Data:	0x020001 // Designate the seesion S2 of inventory, designate the Inv flag A of inventory, antenna 1
Verification code Checksum:	0xCE

Reader returns packet:

If there is a tag response, return the following packet (multiple):

NO.	field	NO.of	fixed	remarks
-----	-------	-------	-------	---------

		bytes	value	
1	Head	1	0xA0	
2	Len	1		
3	Address	1		
4	Cmd	1	0x89	
5.1	Data	Ant	1	Antenna number
5.2		PC	2	The PC of the tag
5.3		EPC	random length	The EPC number of the tag
5.4		RSSI	4	Real-time RSSI of the tag
5.5		Freq	3	Read back the frequency of the tag
6	Checksum	1		

Reference example (for reference only):

Data frame header Head: 0xA0
Packet length Len: 0x19
Reader address Address: 0x00
command code Cmd: 0x89
data packet Data: 0x01 3000 E280689400005016A9874C56 E60AD18D

0DBBA0

//01 Antenna 1
// A PC with 3000 tags
// E200000000004016A9875056 The EPC number of the tag
// E621609A Real-time RSSI of the tag
// Frequency parameters of the 0DBBA0 tag

Verification code Checksum: 0xD2

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x89	
5	Data	1	Error code	0x11: Operation failed 0x22: The antenna is not connected 0x32: Tag read error 0x36: An unoperable tag 0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter 0x43: wordcnt parameter exceeds the specified limit

6	Checksum	1		
---	----------	---	--	--

Reference example (for reference only):

Data frame header Head: 0xA0
 Packet length Len: 0x04
 Reader address Address: 0x00
 command code Cmd: 0x89
 data packet Data: 0x36
 Verification code Checksum: 0x9D

3.46. 8CH—— Stop inventory

Command: cmd_stop_inventory

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x 8C	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 8C D1

The reader successfully executes without returning a command

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x 8C	
5	Data	1	Error code	0x11: Operation failed
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 8C 11 D2

Data analysis: 11: Operation failed

3.47. 8DH —— Select the tag

Command: cmd_set_select

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x8D	
5.1	Select	Enable	1		0: select is not enabled; 1: select is enabled
5.2		SelParam	1		[7:5]Target,[4:2]Action,[1:0]MemBank It is thus constituted The combination of target (3bits) + Action (3bits) + MemBank (2bits) is formed, for example, target is S2 (010), Action is 000, MemBank is EPC (01) Combination 0100 0001 is 41 (Hex)
5.3		Pointer	4		Mask starts at address bit
5.4		MaskLen	1		Mask length
5.5		Truncate	1		0: Truncate is not enabled; 1: Truncate is enabled
5.6		Mask		random length	mask
5	Checksum		1		

Reference example (for reference only):

Data frame header Head: 0xA0
Packet length Len: 0x17
Reader address Address: 0x00
command code Cmd: 0x8D
data packet Data: 0x01 41 00000020 60 01 E200000000004016A9875056

//01 Enable;
//41 SelParam;
//00000020 Mask start address
//60, mask length
//01 Enable Truncate
// E200000000004016A9875056 EPC code

Verification code Checksum: 0xEB

Reader returns packet:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	4	
3	Address		1		
4	Cmd		1	0x 8 D	

5	Data	1		0x10: Operation successful
6	Checksum	1		

Reference example (for reference only):

Data frame header Head: 0xA0
Packet length Len: 0x04
Reader address Address: 0x00
command code Cmd: 0x8D
data packet Data: 0x10
Verification code Checksum: 0xBF

3.48. 8EH—— Get the tag

Command: cmd_get_select

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x 8 E	
6	Checksum	1		

Reference example (for reference only):

Data frame header Head: 0xA0
Packet length Len: 0x03
Reader address Address: 0x00
command code Cmd: 0x8E
Verification code Checksum: 0xCF

Reader returns packet:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x8E	
5.1	Select	Enable	1		0: select is not enabled; 1: select is enabled
5.2		SelParam	1		[7:5]Target,[4:2]Action,[1:0]MemBank It is thus constituted The combination of target (3bits) + Action (3bits) + MemBank (2bits) is formed, for example, target is

					S2 (010), Action is 000, MemBank is EPC (01) Combination 0100 0001 is 41 (Hex)
5.3		Pointer	4		Mask starts at address bit
5.4		MaskLen	1		Mask length
5.5		Truncate	1		0: Truncate is not enabled; 1: Truncate is enabled
5.6		Mask		random length	mask
5	Checksum		1		

Reference example (for reference only):

Data frame header Head: 0xA0
 Packet length Len: 0x17
 Reader address Address: 0x00
 command code Cmd: 0x8E
 data packet Data: 0x01 41 00000020 60 01 E200000000004016A9875056
 //01 Enable;
 //41 SelParam;
 //00000020 Mask start address
 //60, mask length
 //01 Enable Truncate
 // E200000000004016A9875056 EPC code
 Verification code Checksum: 0xEA

3.49. 90H—— Get the inventory buffer

Command: cmd_get_inventory_buffer
 host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x90	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 90 CD

Reader returns packet:

Successful return: This data may return multiple items. The number is equal to the number of tags in the cache (no duplicate data).

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x90	
5.1	Data	InvDataLen	1		The length of the valid data for the tag being operated on. (PC+CRC+EPC). The unit is bytes. If the national standard tag is returned only (PC+EPC)
5.2		InvData	random length		Operate valid data of the tag. The PC (2 bytes) + EPC (according to the tag specification) + CRC (2 bytes) is the entire content in the EPC storage area.) If the national standard tag only returns (PC (2 bytes) + EPC
5.3		RSSI	4		The signal strength when the tag was first read.
5.4		Freq	3		The frequency with which you read the tag for the first time.
5.5		Ant	1		The antenna number read for the first time.
5.6		InvCount	1		The number of successful reads of the tag. If this value is 0xFF, it means that the number of successful reads is greater than or equal to 255.
6	Checksum		1		

Reference example:

Serial port data: A0 1D 00 90 10 30 00 E2 80 69 95 00 00 40 05 CF 27 A4 63 CE 11 E8 19 22 0A 0E 25 18 01 01 78

Data analysis: 10: Valid data length of the tag (PC+EPC+CRC)

30 00: PC

E280699500004005 CF 27 A463: EPC of the tag

CE 11: CRC

E8 19 22 0A: rssi

0E 2518: Frequency of the tag to be read

01: The antenna number read for the first time

01: Number of successful reads

pay attention to:

★ After the command is completed, the data in the cache is not lost and can

be extracted multiple times.

★ If the cmd_inventory command is run again, the tags stored in the disk will be accumulated in the cache.

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x90	
5	Data	1	Error code	0x38: No epc data
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 90 38 94

Data analysis: 38: No epc data

3.50. 91H—— Get and clear the inventory buffer

Command: cmd_get_and_reset_inventory_buffer

Refer to the inventory cache query command for data format.

When the command is successfully completed, all the data in the cache will be cleared.

3.51. 92H—— Get the total number of tag in the inventory buffer

Command: cmd_get_inventory_buffer_tag_count

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x92	
5	Checksum	1		

Reference example (for reference only):

Data frame header Head: 0xA0

Packet length Len: 0x03

Reader address Address: 0x00
 command code Cmd: 0x92
 Verification code Checksum: 0xCB

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	5	
3	Address	1		
4	Cmd	1	0x92	
5	Data	2		Number of tag data in the buffer. No duplicate data
6	Checksum	1		

Reference example (for reference only):

Data frame header Head: 0xA0
 Packet length Len: 0x05
 Reader address Address: 0x00
 command code Cmd: 0x92
 data packet Data: 0x0002
 Verification code Checksum: 0xC7

3.52. 93H—— Reset inventory buffer

Command: cmd_reset_inventory_buffer
 host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0x93	
5	Checksum	1		

Reference example:

Serial port data: A0 03 00 93 CA

Data analysis: None

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
-----	-------	-------------	-------------	---------

1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x93	
5	Data	1	Error code	0x10: Operation successful 0x11: Operation failed
6	Checksum	1		

Reference example:

Serial port data: A0 04 00 93 10 B9

Data analysis: 10: Operation successful

3.53. 95H—— Write with SM7 encryption(optional)

Command: cmd_sm7_write

host computer sends instruction data:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x95	
5.1	Data	Password	4		Enter the password for the tag
5.2		WordCnt Write the length of data	2		WORD (16 bits) length, refer to the tag specification for the value
5.3		WordData	WordCnt *2		The data written (variable length) is twice the length of the data written.
6	Checksum		1		

The reader returns the data packet

Returns the following data packets (multiple), the number of which is equal to the number of tags written (no duplicate data):

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x95	
5.1	Data	TagCount	2		Total number of successfully operated tags. 16 bits.
5.2		DataLen	1		The length of the valid data in the tag

					being operated on. (PC+CRC+EPC). The unit is bytes.
5.3		Write Data	random length		Operate valid data of the tag. PC (2 bytes) + EPC (according to tag specification) + CRC (2 bytes) (PC (2 bytes) + EPC + CRC (2 bytes) is the entire content of the EPC storage area.)
5.4		ErrCode	1		The result of the operation on the tag, i.e. the error code.
5.5		AntID	1		The high 6 bits are the first read frequency parameters, and the low 2 bits are the antenna number.
5.6		Write Count	1		The length of data written to the tag
6	Checksum		1		

Failure return:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x 95	
5	Data	1	Error code	0x11: Operation failed
6	Checksum	1		

3.54. 96H—— Read with SM7 decryption (optional)

Command: cmd_sm7_read

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	0x04	
3	Address	1		
4	Cmd	1	0x 96	
5.	Data	1		The length of the data read. The unit is bytes.
6	Checksum	1		

Reference example (for reference only):

Data frame header Head:	0xA0
Packet length Len:	0x04
Reader address Address:	0x00
command code Cmd:	0x8C
data packet Data:	0x11
Verification code Checksum:	0xD2

The reader returns the data packet

Returns the following data packets (multiple) in the number equal to the number of tags read (no duplicate data):

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x 96	
5.1	Data	TagCount	2		Total number of successful operation tags. 16 bits.
5.2		DataLen	1		The effective data length of the tag being operated on (PC+CRC+EPC+tag data read). The unit is bytes.
5.3		Read Data	random length		Valid data for the tag being operated. PC (2 bytes) + EPC (according to tag specification) + CRC (2 bytes) + read data. (PC (2 bytes) + EPC + CRC (2 bytes) is the entire content of the EPC storage area.)
5.4		ReadLen	2		Read the length of data to be read. The unit is bytes.
5.5		AntID	1		The high 6 bits are the first read frequency parameters, and the low 2 bits are the antenna number.
5.6		ReadCount	1		The number of times the tag has been successfully operated.
6	Checksum		1		

Failure return:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	4	
3	Address		1		
4	Cmd		1	0x 96	

5	Data	1	Error code	0x11: Operation failed
6	Checksum	1		

3.55. 97H——SM7 PK update (optional)

Command: cmd_sm7_pk_update

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	0x1 3	
3	Address	1		
4	Cmd	1	0x9 7	
5.1	Data	16		SM7 encryption key 128bit
6	Checksum	1		

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	4	
3	Address	1		
4	Cmd	1	0x97	
5	Data	1	Error code	0x10: Operation successful 0x11: Operation failed
6	Checksum	1		

3.56. 98H—— Mul-Sauth for Fudan [Micro tags](#)

Command: cmd_gb_mul_SeU_Auth

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	0x13	
3	Address	1		
4	Cmd	1	0x 98	

5.1	Data	Password	16		Two-way authentication key
6	Checksum		1		

The reader returns the data packet

Returns the following data packets (multiple) in the number equal to the number of tags destroyed (no duplicate data):

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1		
3	Address		1		
4	Cmd		1	0x 49	
5.1	Data	TagCount	2		Total number of successfully operated tags. 16 bits.
5.2		DataLen	1		The effective data length of the tag being operated on (PC +EPC). The unit is bytes.
5.3		EPC	random length		Operate valid data of the tag. PC (2 bytes) + EPC (according to tag specification)
5.4		ErrCode	1		The result of the operation on the tag, i.e., the error code.
5.5		AntID	1		The high 6 bits are the first read frequency parameters, and the low 2 bits are the antenna number.
6	Checksum		1		

Failure return:

NO.	field		NO.of bytes	fixed value	remarks
1	Head		1	0xA0	
2	Len		1	4	
3	Address		1		
4	Cmd		1	0x 49	
5	Data		1	Error code	0x11: Operation failed 0x58: Two-way authentication failed 0x59: Two-way authentication succeeded 0x36: An unoperable tag 0x37: Successful inventory but access failed 0x40: Access tag error or password error 0x41: Invalid parameter
					0x60: Insufficient power supply for the tag 0x61: Insufficient tag permissions 0x62: Memory address is out of range 0x63: Memory is locked

6	Checksum	1		
---	----------	---	--	--

3.57. The E1H—— Warning of an Overheating

Data packets reported by the reader:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	3	
3	Address	1		
4	Cmd	1	0xE1	
5	Checksum	1		

3.58. F3 H—— Read the UUID

Command: cmd_read_uuid

host computer sends instruction data:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	0x3	
3	Address	1		
4	Cmd	1	0xF3	
5	Checksum	1		

Reader returns packet:

NO.	field	NO.of bytes	fixed value	remarks
1	Head	1	0xA0	
2	Len	1	0x13	
3	Address	1		
4	Cmd	1	0xF3	
5	Data	16	uuid	128bit
6	Checksum	1		

Reference example (for reference only):

Data frame header Head:	0xA0
Packet length Len:	0x03
Reader address Address:	0x00
command code Cmd:	0xE1
Verification code Checksum:	0x7C

4. Error code table

NO.	price	name	explain
1	0x10	command_success	The command was successfully completed
2	0x11	command_fail	The command execution failed
3	0x12	Custom inventory complete	Customized inventory is complete
4	0x13	Fast switch antenna inventory complete	Multiple antenna polling is complete
5	0x20	mcu_reset_error	CPU reset error
6	0x21	cw_on_error	Open CW error
7	0x22	antenna_missing_error	The antenna is not connected
8	0x23	write_flash_error	Write Flash error
9	0x24	read_flash_error	Read Flash error
10	0x25	set_output_power_error	The transmitting power is set incorrectly
11	0x31	tag_inventory_error	Inventory tag error
12	0x32	tag_read_error	Read the tag incorrectly
13	0x33	tag_write_error	tagging error
14	0x34	tag_lock_error	The lock tag is incorrect
15	0x35	tag_kill_error	The inactivated tag is incorrect
16	0x36	no_tag_error	The tag is not operable
17	0x37	inventory_ok_but_access_fail	Successful inventory but access failed
18	0x40	access_or_password_error	The access tag is incorrect or the access password is incorrect
19	0x41	parameter_invalid	Invalid parameter
20	0x42	parameter_invalid_wordCnt_too_long	wordCnt The parameter

			exceeds the specified length
21	0x43	parameter_invalid_membank_out_of_range	MemBank Parameter exceeds the range
22	0x44	parameter_invalid_lock_region_out_of_range	The Lock data area parameter is out of range
23	0x45	parameter_invalid_lock_action_out_of_range	LockType Parameter exceeds the range
24	0x46	parameter_reader_address_invalid	The reader address is invalid
25	0x47	parameter_invalid_antenna_id_out_of_range	Antenna_id exceeds the range
26	0x48	parameter_invalid_output_power_out_of_range	The output power parameter is out of range
27	0x49	parameter_invalid_frequency_region_out_of_range	The RF specification area parameter is out of range
28	0x4A	parameter_invalid_baudrate_out_of_range	The 波特率参数超出范围.
29	0x4C	parameter_epc_match_len_too_long	The EPC matching length exceeds the limit
30	0x4D	parameter_epc_match_len_error	The EPC matching length is incorrect
31	0x4E	parameter_invalid_epc_match_mode	The EPC matching parameters are out of range
32	0x4F	parameter_invalid_frequency_range	The frequency range setting parameter is incorrect
33	0x50	fail_to_get_RN16_from_tag	The RN16 cannot receive the tag
34	0x53	rf_chip_fail_to_response	The RF chip is not responsive
35	0x54	fail_to_achieve_desired_output_power	The output does not reach the specified output power
36	0x55	copyright_authentication_fail	The copyright certification was not passed
37	0x56	spectrum_regulation_error	The spectrum specification is set incorrectly

38	0x57	output_power_too_low	Output power is too low
	0x58	GB sm7 double identify failed	The national standard bidirectional authentication failed
	0x59	GB sm7 double identify success	The national standard bidirectional authentication is successful
	0x60	GB tag be short of power	The tag power is insufficient
	0x61	GB tag permission error	Insufficient authority
	0x62	GB tag memory over limit	Memory area exceeded
	0x63	GB tag memory has be locked	Memory is locked
	0x64	GB tag password error	wrong password
	0x65	GB identify error	Authentication failed
	0x66	GB unknown error	Unknown error

5. Configuration table for frequency parameter

frequency parameter	Corresponding frequency	frequency parameter	Corresponding frequency
0 (0x00)	865.00 MHz	30 (0x1E)	913.50 MHz
1 (0x01)	865.50 MHz	31 (0x1F)	914.00 MHz
2 (0x02)	866.00 MHz	32 (0x20)	914.50 MHz
3 (0x03)	866.50 MHz	33 (0x21)	915.00 MHz
4 (0x04)	867.00 MHz	34 (0x22)	915.50 MHz
5 (0x05)	867.50 MHz	35 (0x23)	916.00 MHz
6 (0x06)	868.00 MHz	36 (0x24)	916.50 MHz
7 (0x07)	902.00 MHz	37 (0x25)	917.00 MHz
8 (0x08)	902.50 MHz	38 (0x26)	917.50 MHz
9 (0x09)	903.00 MHz	39 (0x27)	918.00 MHz
10 (0x0A)	903.50 MHz	40 (0x28)	918.50 MHz
11 (0x0B)	904.00 MHz	41 (0x29)	919.00 MHz
12 (0x0C)	904.50 MHz	42 (0x2A)	919.50 MHz
13 (0x0D)	905.00 MHz	43 (0x2B)	920.00 MHz
14 (0x0E)	905.50 MHz	44 (0x2C)	920.50 MHz
15 (0x0F)	906.00 MHz	45 (0x2D)	921.00 MHz
16 (0x10)	906.50 MHz	46 (0x2E)	921.50 MHz
17 (0x11)	907.00 MHz	47 (0x2F)	922.00 MHz
18 (0x12)	907.50 MHz	48 (0x30)	922.50 MHz
19 (0x13)	908.00 MHz	49 (0x31)	923.00 MHz
20 (0x14)	908.50 MHz	50 (0x32)	923.50 MHz
21 (0x15)	909.00 MHz	51 (0x33)	924.00 MHz
22 (0x16)	909.50 MHz	52 (0x34)	924.50 MHz
23 (0x17)	910.00 MHz	53 (0x35)	925.00 MHz
24 (0x18)	910.50 MHz	54 (0x36)	925.50 MHz
25 (0x19)	911.00 MHz	55 (0x37)	926.00 MHz
26 (0x1A)	911.50 MHz	56 (0x38)	926.50 MHz
27 (0x1B)	912.00 MHz	57 (0x39)	927.00 MHz
28 (0x1C)	912.50 MHz	58 (0x3A)	927.50 MHz
29 (0x1D)	913.00 MHz	59 (0x3B)	928.00 MHz

6. RSSI parameter calculation method

(C language description)

```
const uint8_t para_B[5][8] = {
    {43,43,45,49,43,43,45,49},
    {43,43,45,49,43,43,45,49},
    {43,43,45,49,43,43,45,49},
    {53,53,48,43,49,45,43,43},
    {47,47,47,47,46,43,43,43}
};

const int para_C[5][8] = {
    {43,43,45,49,43,43,45,49},
    {43,43,45,49,43,43,45,49},
    {43,43,45,49,43,43,45,49},
    {-283,-283,-283,-283,-283,-283,-283,-283},
    {-303,-283,-253,-238,-304,-313,-280,-266}
};

int Calculate_Rssi(char data[],uint8_t epc_len){
    uint8_t rssi_mode = 0, hardware_mode = 0;
    int B=0,C=0,D=0, RssiVal = 0;
    float A = 1.0f, rssi_temp = 0.0f;
    union{
        uint32_t u32;
        uint8_t chr[4];
    }UNION;
    if(epc_len == 0)epc_len = 1;
    rssi_mode      = (data[0]&0xE0) >> 5;
    hardware_mode  = (data[0]&0x1E) >> 1;
    UNION.chr[3] = data[0] & 0x01;
    UNION.chr[2] = data[1];
    UNION.chr[1] = data[2];
    UNION.chr[0] = data[3];
    B = para_B[hardware_mode][rssi_mode];
    C = para_C[hardware_mode][rssi_mode];
    rssi_temp = (UNION.u32/epc_len)*A;
    RssiVal = (B * log10(rssi_temp)) + C + D;
    if(RssiVal > 0)RssiVal = 0;
    else if(RssiVal < -90)RssiVal = -90;
    return RssiVal;
}
```

7. Checksum calculation method

(C language description)

```
uint8_t CheckSum(uint8_t *uBuff, uint8_t uBuffLen)
{
    unsigned char i, uSum = 0;
    for (i = 0; i < uBuffLen; i++)
    {
        uSum = uSum + uBuff[i];
    }
    uSum = (~uSum) + 1;
    return uSum;
}
```

8. Default factory configuration

Parameter name	Parameter value (unit)	remarks
Baud rate	115200 (bps)	The baud rate of the command communication serial port can be configured, but the baud rate of the serial port for printing logs cannot be configured (115200bps)
Reader address	0x00	The public address is used by default at the time of delivery
transmitting power	MU601:20 (dBm) MU606:30 (dBm) MU608:33 (dBm)	
Work antenna	1	
work pattern	frequency hopping	
Working frequency band	FCC	902mHz-928mHz
Number of frequency points	53	Use this parameter when hopping
Frequency spacing	500 (KHz)	Use this parameter when hopping
Starting frequency	902 (MHz)	
Communication bandwidth	Miller 200KHz 25us	
buzzer	1 (Open)	